

Week-6 Essay (Chapter 8, 9 of WMD and Chapter 6 of Everybody Lies)

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Table of Contents

Introduction.....	3
Literature Review.....	4
Analysis of the Issue	4
Conclusion and Limitations	6
References.....	6

Introduction

In chapter 8 of "Weapons of Math Destruction" by Cathy O'Neil explores how big data and algorithms in finance can make inequality worse by reinforcing existing biases. Cathy O'Neil explains that these systems, built to be efficient, often care more about making money than being fair. As a result, they can hurt low-income people and minorities more than others. Chapter 9 discusses how predictive algorithms used in the insurance industry are transforming the sector from its traditional purpose of communal risk-sharing into one driven by hyper-personalized pricing and profit maximization. Cathy O'Neil explores how these algorithms, while efficient, create systemic inequalities by penalizing the most vulnerable members of society.

In Chapter 6 of *Everybody Lies*, titled as All the World's a Lab by Seth Stephens-Davidowitz explores how the world is like a giant testing lab thanks to big data. It explains how A/B testing, a method used to make better decisions and improve things, can help us understand human behavior. With all the data we leave online, it's easier than ever to learn about people's preferences, feelings, and social patterns. These digital traces give us insights into things we didn't know before.

Literature Review

Chapter 8 of *Weapons of Math Destruction* explains how credit scoring algorithms, like FICO scores, worsen inequality and harm marginalized communities. These systems use factors like zip codes and spending habits, which often reflect past biases, especially regarding race and income. For example, people from low-income neighborhoods may have lower credit scores due to where they live, making it harder for them to get credit and improve their financial situation. Cathy O'Neil uses the story of Taylor, a fictional character, to show how these systems fail by ignoring personal struggles. Payday loan companies exploit these flaws, trapping vulnerable people in high-interest debt. The chapter also points out that these algorithms are hard to understand, leaving people powerless to challenge their scores, locking them out of opportunities while wealthier people benefit.

Chapter 9 examines the history of insurance, starting with actuarial science, which aimed to spread risk across a population. Over time, with the rise of big data and predictive analytics, insurance companies shifted from community-based pricing to individual risk profiling. These algorithms now use unrelated data, like credit scores or shopping habits, to maximize profits. O'Neil highlights discriminatory practices in auto insurance, such as charging a driver with a clean record but poor credit higher premiums than a wealthier driver with a history of reckless driving. She criticizes companies like Allstate for using "price optimization algorithms" that unfairly burden low-income groups. The chapter also draws parallels to redlining in the housing market, where discrimination based on race and income still exists today through these modern algorithms.

Chapter 6 of *Everybody Lies* explores two main types of experimentation: A/B testing and natural experiments. A/B testing involves making small changes to one variable at a time and measuring its impact on outcomes. Companies like Google, Amazon, and Netflix use this method to improve user experience and content delivery. On the other hand, natural experiments happen when real-life events or uncontrollable factors create situations that mimic controlled experiments. Examples include differences in geography, timing, or policies, which can serve as groups for analysis. Stephens-Davidowitz highlights that while these methods are not new, they have become much more powerful due to the vast amount of data now available. Unlike previous studies that relied on small samples or artificial settings, these new methods can capture a much deeper understanding of human behavior by examining real-world scenarios on a larger scale.

Analysis of the Issue

The chapter 8 looks at the serious problems with algorithmic credit scoring systems and how they make things worse for people already at a disadvantage. One major issue is that these systems tend to favor wealthy individuals who get more personalized attention, while poorer people are stuck with automated processes that don't consider their unique situations. This means their financial struggles are treated as simple numbers without understanding the full story. Payday lenders take advantage of this by using these biased algorithms to target vulnerable people with high-interest loans, trapping them in a cycle of debt. O'Neil also points out that these systems don't have ways to improve or learn from their mistakes. Instead of getting better over time, they just keep reinforcing unfair practices, making it even harder for people to break free from the system.

In chapter 9 O'Neil discusses how algorithms in insurance lead to unfair pricing, with some customers getting discounts while others are overcharged based on wealth or education, not actual risk. High premiums push vulnerable people into risky financial behaviors, making their situation worse. The idea of insurance as a shared safety net has faded, as people now pay based on their own risks. She also criticizes the lack of transparency in these systems, leaving customers unable to question unfair rates. In the trucking industry, surveillance tech tracks drivers' movements, but it can be misused to blame them or deny claims, showing the darker side of technology.

The chapter 6 provides interesting examples of how big data and real-life experiments reveal surprising insights. In political campaigns, A/B testing is used to figure out which messages connect best with voters. Even small changes in words or images can have a big impact on donations and voter turnout. In the music world, natural experiments have shown how social influence affects musical preferences. For example, when people saw which songs others were downloading, some tracks became much more popular, proving how social validation shapes choices. In housing markets, an online experiment found that listings with racially ambiguous names got fewer responses than those with traditionally white names, showing evidence of implicit bias. These examples show how A/B testing and natural experiments can uncover important truths about human behavior, often revealing things we didn't expect.

Conclusion and Limitations

In chapters 6, 8, and 9 of *Weapons of Math Destruction* and *Everybody Lies*, the authors discuss how algorithms impact society. Credit-scoring models widen inequality by hiding biases, and insurance algorithms often prioritize profit over fairness, hurting vulnerable groups. O’Neil calls for more transparency and regulation to ensure these systems benefit everyone. Meanwhile, *Everybody Lies* shows how platforms like Google, Facebook, and Amazon reveal more accurate insights into human behavior than surveys. In my view, while big data offers valuable insights, it’s crucial to ensure fairness and accountability to prevent harm and prioritize people’s well-being.

References

- Davidowitz, S. S. (2017). *Everybody lies: Big data, new data, and what the internet can tell us about who we really are*. Dey Street Books.
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